Mathematics That Means Something

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1 Analysis

Let us collect classical results. The Gaussian integral

$$\int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x = \sqrt{\pi} \tag{1}$$

underlies the normal distribution. Euler's identity reads $e^{i\pi} + 1 = 0$, and the Basel problem asks for

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$$

Let $f\colon \mathbb{R} \to \mathbb{R}$ be continuous.

Theorem 1 (Mean value)**.** *If f is differentiable on (a,b) and continuous on $[a,b]$, then there is a $c \in (a,b)$ with*

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof. The function $g(x) = f(x) - ((f(b)-f(a))/(b-a))(x-a)$ vanishes at a and at b, so Rolle's theorem applies to it. ∎

The limit

$$\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e$$

defines e, and for $|x| < 1$ the geometric series gives $\sum_{k=0}^{\infty} x^k = 1/(1-x)$.

2 Algebra

Definition 1. A matrix $A \in \mathbb{R}^{n\times n}$ is *orthogonal* when $A^{T}A = I$.

Determinants of small matrices are written out directly:

$$\det\begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}$$

The binomial theorem states

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k} \tag{1}$$

and the quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

3 Probability

For a random variable X with finite variance,

$$\begin{aligned} \operatorname{Var}(X) &= \mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 \end{aligned}$$

and Chebyshev's inequality bounds the tails:

$$\Pr(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2} \quad \text{for all} \quad k > 0$$

The absolute value is

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{otherwise} \end{cases}$$

Corollary (1) with $x = y = 1$ gives $\sum_{k=0}^{n} \binom{n}{k} = 2^n$; compare (1) and Theorem 1.